Proposed Risk Management Decision Support Methodology for Oil and Gas Construction Projects

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Abstract Oil and gas construction projects are complex and risky due to their dynamic nature and environment, as they involve a considerable number of stakeholders. The contracting companies in Iraq find many challenges when managing the risky events in an environment that categorized with poor supplies, security threats, unskilled workforce and logistics difficulties. Evidence in the literature indicates that there is a lack of unified methodology for project management and especially for risk management in Iraq. Also, the rise in the global energy demand increases the need for a workable, effective and efficient risk management methodology for such projects. Thus, the purpose of this research is to develop an integrated decision support methodology for managing the risk factors in oil and gas construction projects in Iraq. The proposed methodology has been developed to support the local and international contracting companies working in Iraqi oil and gas fields when making decisions regarding the risk factors during the project life cycle. The proposed methodology consists of the following phases: risk planning, risk identification using documentation review and expert interviews, risk analysis using a multi-criteria risk analysis model based on fuzzy set theory, risk effect prediction on project time and cost using artificial neural network (ANN), selection of risk response actions using Gravitational Search Algorithm (GSA) optimization technique and finally, designing an integrated web-based risk management decision support platform. The adopted methodology will enable decision makers to assess the oil and gas projects risky events, support their decision during planning and work implementation stages, gain experience in risk management through exercising and implementing risk management on scientific and documented bases; organize and document the knowledge-based information for decision makers.

Keywords Oil and gas · Project management · Risk management · Decision support system

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1 Introduction

Energy demand is rising, and it is expected that global investment in the energy-related sectors will reach 38 trillion through 2035 (Accenture 2012; Jayaraman et al. 2015). Oil and Gas industry considered to be as a backbone of any country's economy (Mubin and Ghaffar 2008). All construction projects are usually confronted by many types of risk factors. There is always some sort of uncertainty in meeting construction project objectives (time, cost and quality). These objectives are risky due to several factors such as: the complexity of the construction projects and the unique nature that makes them scientifically different from other similar projects (Kendrick 2015; Seddeeq et al. 2019; Salas 2014). The construction risks facing oil and gas construction projects can be social, political, financial, and technical in nature, as well as related to the natural disasters and natural environment (Aven et al. 2007; Zhao et al. 2012). Recognizing and Understanding the risk factors associate with energy sector is not only a construction market need, but also it is essential to the upstream and downstream oil and gas industry sub-sectors (De Maere d'Aertrycke et al. 2017). Construction projects at the oil and gas fields play a significant role in the sustainable supply chain and processing (Hong et al. 2016). Because to this importance, governments of developing countries are under pressure to ensure that construction projects will be completed on time within the estimated budgets and least amount of uncertainties and risks (Dehdasht et al. 2015). The continues need to ensure that construction projects are managed effectively led both of the academia and industry emphasising on the development of reliable methodologies for managing projects risks (Elhoush and Kalafungin 2017). Project risk management (PRM) has been known as a strong factor in need of careful assessment to ensure project risks are minimized (Carvalho and Rabechini 2015; Salazar-Aramayo et al. 2013). Unfortunately, the construction industry is not as advanced at assessing and analysing risk factors in comparison with other industries (Laryea 2008; Taroun 2008). Lack of risk management methodology during oil and gas construction projects will not always leads to cost overrun and delays but it also can be crucial for pricing the energy policies (AL-Zwainy et al. 2016). The challenge is when the project moves from design stage into site production stage. The management team are so stretched with everyday challenges that they often fail to keep the risk register up to date and fail to recognize the interdependence of the risky events. It is therefore the aim of this research is to propose a risk management methodology for supporting decision makers when managing the construction risks at the oil and gas field in Iraq. This theoretical work illustrates the integration of risk management phases and analytical approaches for supporting risk decision making.

2 Literature Review

2.1 Oil and Gas Industry in Iraq

Iraq has become the 4th largest oil producer in the world after Russia, Saudi Arabia and USA (World Bank 2017) and the 2nd largest oil producer in OPEC (EIA 2017) with a total oil production of 4.5 million bpd in 2016 (BP 2017). Iraq considered to be one of the few regions left where vast oil reserves, proven and unknown, have hardly discovered (Donovan 2013). The Oil reserves of Iraq are enormous, it considered to be the 5nd largest reserves in the World with around 142.8 Billion bpd. Iraq natural gas production reached 189.4 billion m^3 in 2015 with a total proven reserve of 3.7 trillion m^3 for the same year (BP 2017). The energy sector in Iraq is heavily depends on petroleum production (Maniruzzaman and AL-Saleem 2017), with around (90%) of its energy requirements met with petroleum (Balanchard 2018). Iraqi oil and gas industry contribute for most of the Gross Domestic Product (GDP), foreign exchange earnings and public revenues (IEITI 2015). It is, therefore, central to the country finical position and critical to the vitality of the country economy and the ongoing reconstruction efforts of the country, particularly with regard to oil, gas infrastructure and development (IEITI 2015). According to the World Bank report in 2017, the Oil and Gas fields in Iraq contributes to around 65% of its gross domestic product (GPD), and over 90% of the public income and most of its foreign exchange revenue (World bank group 2018), while the non-oil sector non-productive as the oil and gas sector due to weak service delivery, corruption and insecurity situation (Balanchard 2018). Iraq has (22) major oil and gas fields separated mainly in the south, middle and north.

2.2 Project Risk Management

In the construction industry risk is defines as a variable, condition or uncertain event that if occurred it will have an impact on at least one of the project objectives (Jin et al. 2017). Numbers of researchers dated the origin of the modern risk management to be between the 1950s and 1960s (Crockford 1982; Arthur and Richard 1995; Scott and Gregory 2003). The first two scientific books were published by Mehr and Hedges (1963) and Scott and Gregory (1964) were they covered pure risk management fundamentals and principals. Project risk management was practised in the mid of 1980s (Rezakhani 2012), it falls inside the nine knowledge areas of the project management institute's project management body of knowledge (PMI 2008). Risk management is defined as the proactive, constant and organized system to manage, realize and communicate risk events (Hutchins 2016). Risk management in construction projects usually has two main goals: to avoid threats and to exploit the opportunities (Smith et al. 2013). An effective risk management depends on previous historical data then

looking forward to thorough planning to narrow future risks area where future failures can be avoided (Kendrick 2015). Effective management of risks does not only concentrate on minimizing risk consequences, but it gives sufficient support to the activities that adopt innovation, so greater benefits can be achieved. According to the authors, risk management cannot be utilized as a tool to predict the future. Instead, it facilitates the project to make proper decisions depending on related information in which the decision that depends on insufficient information can be neglected in which it will lead to efficient project performance. Risk management assures that almost all project problems are discovered in the early project stages without overpassing the project budget or missing the project schedule (Tamak and Bindal 2013). Understanding the risks confronting a construction project helps the project team to meet the project parties' expectations more effectively. The assessment of these risks withdraws the decision makers attention to the difficulties that might be faced in order to fulfil the project objectives and the enterprise overall expectations (WSDOT 2014). Therefore, risk management is considered as an integration of decision making and good management at all levels of the organization (Berg 2010). The proper implementation of risk management aimed at providing sensible assurance as to the successful accomplishment of the project objectives and helps the company financial goals. The Key benefits of applying systematic risk management can be summarized below as it described by Dinu (2012), Siang and Ali (2012), and Shad and Lain (2015). It reduces threats and increases their opportunities, so it maximizes the probability of accomplishing objectives, it provides effective management that maximizes the work effectiveness and efficiency by minimizing stress and waste, it helps in providing proactive management of threats which have an important effect on the whole project objectives, it fills strategic gaps by shaping and updating the risk culture at all management level, it decreases the instability of the construction activities and discourages the acceptance of financially unsound projects and it protects credibility and reputation of the company.

3 Research Problem

Studies about risk management have been presented since the 1950s (Crockford 1982; Arthur and Richard 1995; Scott and Gregory 2003). Risk management process steps and characteristics based on several risk management international and professional standards such as: British Standards Institution (BS6079-3: 2000), Institute of Risk Management (IRM: 2002), Standards Australia/Standards New Zealand (AS/NZS 4360: 2004), the international organization for standards (ISO 31000: 2009), International Electrotechnical Commission (IEC 31010: 2009), Canadian Standards Association (CAN/CSA Q850: 2009), National Institute of Standards and Technology (NIST800-30: 2012), Project management body of knowledge project management institute (PMBOK-PMI: 2013), American Management Association Handbook (AMA: 2014) which are sharing the same crucial steps despite the existence of some differences in some of risk management steps details. These steps are

risk planning, risk identification, risk analysis, risk response and risk monitoring and controlling. However, the risk related information gathered from any project phase is different from the other phases, in another meaning, in each phase, the project has its risk characteristics. Also, the extent of how the decision makers assess the relevant risks becomes different at each phase of the project life cycle. In addition, risk factors of oil and gas construction projects are complex, interrelated and varied. Therefore, the traditional approaches of risk management have limitations and barriers of applicability to such complex projects. The construction projects in Iraq have always been plagued with implementation problems and weak outcomes (Abdulsattar 2017). Iraqi contracting companies find a big number of challenges to mitigate risk events in an environment with poor supplies, security threats, unskilled workforce and logistics difficulties. Also, there were some downsides due to the unstable country economy and political issues. Besides, there is a lack of the adoption of a unified methodology for project management in the Iraqi construction industry (AL-Zwainy et al. 2016) and lack of information and data regarding the construction risk management processes in Iraq.

Based on the issues stated above, there is a need to develop a risk management decision support system methodology for oil and gas construction projects in Iraq.

4 Research Methodology

The nature of the study takes up a qualitative approach as it based on literature review and documents analysis which used for knowledge acquisition to develop a formalised risk management methodology. The investigation began by reviewing the related literature to determine the gaps in the relevant literature and to establish the proposed system requirement in which it helped to identify the most effective techniques to achieve the research aim. Documentation analysis in this research includes risk data, risk behaviour, causes of project failure and adopted strategies for decision making when managing risks at the oil and gas construction project for the local and international oil and gas companies working in Iraqi oil and gas sector.

5 Proposed Risk Management Decision Support Methodology

The proposed risk management decision support methodology is defined by six phases which can be illustrated by Fig. 1.

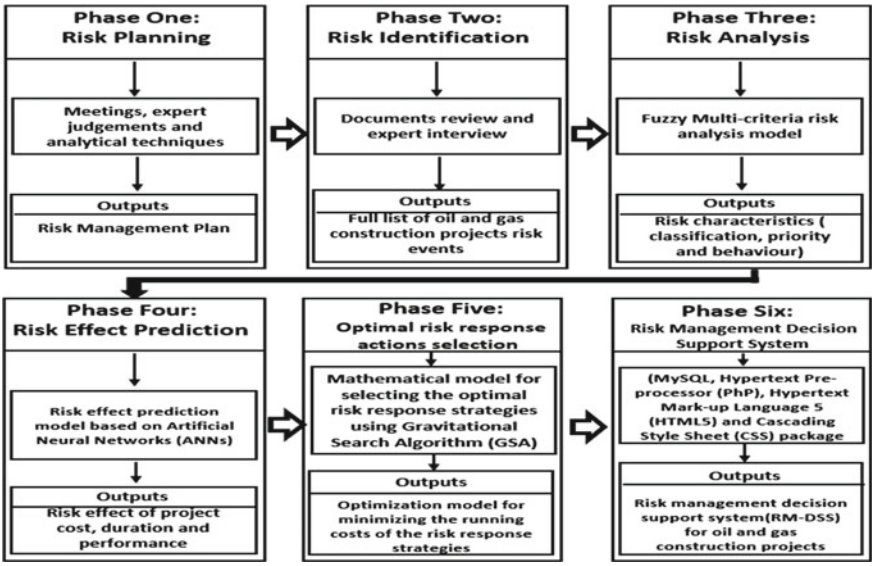

Fig. 1 The proposed risk management decision support system structure

5.1 Phase One: Risk Management Planning

Risk management planning is a systematic approach of deciding how to plan and implement the risk management activities throughout the project life cycle. The development of risk management plan is considered to be one of the most important activities in the risk management process, because it establishes and organize the activities of the project management team when dealing with project risks and uncertainties.

In this Phase, the operations that need to be accomplished by the risk management team are described in detail. It includes defining and scheduling all the activities and procedures needed to observe, evaluate and document the risk factors associated with the project. The outcome of this step is known as risk management plan which has an essential role in (Burek 2007; Nicholas and Steyn 2011): updating the project risk management strategy, finding out the most effective approaches to implement the risk management strategy and nominating the appropriate parties to get the job done. In order to build up a successful risk management plan, enough data must be collected through the project work breakdown structure (WBS) in addition to the project performance standards, cost plan and schedule. Previous effective strategies in similar achieved projects must also be present. Risk management planning activities can be summarized into the following: identifying appropriate cost-risk team, determining the workshops types and durations, establishing risk communication approach, determining the risk identification methods depending on project characteristics, determining the level of risk assessment, scheduling risk assessment

activities, implementation of risk treatment actions, initiating monitoring for risk treatment actions, updating risk management plan and performing post mitigation analysis and report.

5.2 Phase Two: Risk Identification

Risk identification is considered as the most significant step in the risk management process where the risk assessment and response actions cannot be applied only on the risks that been identified and recognized (Burek 2007). According to Giannakos and Louis (2011), risk identification is the essential step for detecting the risk factors that affecting the project objectives. Any failure in identifying any risk factor will lead to the ill implementation of the whole process, which can in turn critically affect the organization's resources. It is also considered as an essential step in constructing all risks profiles. Risk identification is said to be of two phases. The first is the initial identification phase which is usually conducted by firms that have not already identified their risk factors in a structured way or by new firms or projects. The second is the continual risk identification phase which aims at identifying new risk factors never occurred before (Ana-Maria and Doina 2012). Risk identification, classification and risk breakdown structure helps to comprehensively identify all risk. Risk Identification techniques in the proposed methodology are documentation review and expert interview. In order to collect the primary data, two techniques will be carried out: documents review and expert interviews with construction experts.

5.2.1 Documents Review

A traditional and basic technique that involves reviewing reports, drawings, plans, estimates, schedules contracts and other stored information in either hard copy or electronic formats in which the pre-identified opportunities and threats are investigated for signs of potential risk situations. According to Myers (2009), the analysis of documentary data and records is important to identify statements made by key persons in an organization which can be utilized as inputs to the interview guides. The causes of projects failures will be analysed from published reports produced by construction companies working at Iraqi's oil and gas fields which are located in Basra governorate. These report record information such as into risk status records, project sum, project duration, time extensions, variation orders project failure, causes of project failure and their response actions.

5.2.2 Expert Interviews

It is a very effective tool of obtaining rich data from the participants which is crucial for explaining or exploring phenomena or complicated situations. Based on Knight

and Ruddock (2008), a good interview is the art of exploring the beliefs and opinions of the participants and the subject knowledge. This technique is known as a resource-intensive technique because of the organizational time and resource requirement. Interviews are conducted with project managers, contractors, consultants and key stakeholders, where it could be conducted one-on-one or within a group setting which can be utilized to discuss existing concerns, identify any risky event and recommend what measures need to be taken when dealing with such risks. In order to make the interviews with the key personnel more efficient, a set of questions need to be prepared prior to the interviews. Besides being an effective technique for risk identification, the challenges come when interpreting the information obtained from the interviews and to classify the developed risky event properly. In order to obtain the proposed system requirement, interviews with experts are needed; to identify risk factors in oil and gas projects where no documentary data are available; to investigate the risk management current practice, adopted methodologies and challenges in the Iraqi oil and gas construction projects and its impacts on the project outcome; to collect some information needed for the design of the questionnaire surveys such as risk analysis criteria and the effective response strategies.

5.3 Phase Three: Risk Analysis

Risk analysis is the process of defining, analysing and prioritizing the risk factors based on their importance (El-Sayegh and Mansour 2015). The objective of risk analysis is to have a clear understanding of the risk circumstances and consequences then to sort risks according priorities based on a definite risk management strategy. Risk analysis aims at realizing the significance of each risk and the appropriate management path needed. The assessment process should be performed periodically to identify and risky event that should have positive or negative impact on the project objectives (time, cost and quality). The selection of the analysis approach is mainly depending on the level of risk, management experience and project level. Based on that a multi-criteria risk analysis model using fuzzy set theory is proposed to carry out the analysis phase of the identified risks. A multi-criteria risk analysis model based on fuzzy set theory and fuzzy analytic hierarchy method is proposed in this research through literature review and expert interviews. The model intends to achieve systematic and structured analyses of projects risks. According to Muhammad and Mhdhav (2016) and Roghanian and Mojibian (2015); Fuzzy logic is an effective management technique for risk management and for dealing with uncertainties in human decision making. It is an appropriate tool to deal with subjective judgement (Nieto-Morote and Ruz-Vila 2011). It can be employed to develop a model on the base of qualitative data (expert judgment) or quantitative values (Subramanyan et al. 2012). Fuzzy logic formulation and computation is applicable to a large number of engineering fields which ranging from risk assessment to risk pricing algorithm (EL Khalek et al. 2016). The development of the risk analysing model can be achieved through establishing the risk analysing criteria of the proposed model by analysing the limitation

of the traditional multi-criteria analysing models and by taking into consideration the expert interviews outcomes regarding the effective analysing criteria, evaluating the fuzzy language set which depends on the risk analysing criteria, evaluating the failure mode by a triangular fuzzy number depending on the fuzzy set, determining the weights of the proposed criteria model using the fuzzy analytic hierarchy Process and determining each risk rank and importance. The outcomes of this phase are prioritized risk parameters obtained from conducting the questionnaire survey and processed using the proposed fuzzy multi-analysing criteria.

5.4 Phase Four: Risk Effect Prediction

This phase involves the development of a risk effect prediction model based on Artificial Neural Networks (ANNs) which can predict the effect of each risk factor on the project cost and duration. According to Lazarevska et al. (2012), ANNs are a computational mechanism which can analyses, and process data obtained from numerical analysis or experiments. ANNs has been usefully applied for cost estimation and construction scheduling. It is can be successfully utilized to solve problems related to cost estimation, decision making, optimizations and schedule predications in the field of construction engineering and management (Waziri et al. 2017). In this research ANNs will be utilized to determine and predict the duration needed to overcome each risk factor based on the project duration and risk importance obtained from risk analysis phase by Fuzzy set theory and also to predict the total cost needed to treat each risk factor based on the project total cost and each risk importance based on the fuzzy analysis phase in which it will allow a reasonable logical assessment away from randomization when determining the project schedule activities (duration contingencies) and cost of contingencies.

Based on Sang et al. (2014), the Steps for applying ANNs can be summarized as: identifying inputs variables, data collection and encoding, determining the output variables, Ann's architecture design, ANNs training and testing and ANNs validating. Despite the existence of several methods for estimating and predicting the risk effect on project time and cost, such as Analytic Hierarchy Process (AHP), Monte Carlo simulation and expert evaluation, these methods are subjected to personal judgement in which it may led to inaccurate prediction of the failure effect which will have an impact on the risk management decision making. Also, the traditional risk effect deterministic methods may neglect the factors that might have a significant influence on expected prediction values. However, artificial neural networks have been shown ability of solving different prediction problems in construction management field. In this research a back propagation neural network which based on back ward propagation of errors algorithm will be adopted. It considered being one of the most common types of neural networks and the most widely used especially in engineering applications.

The neurons or the nodes activation functions in this type of networks are similar to each other in each layer. The system requirement is represented by two sets of

variables. In order to build the prediction model, two types of variables are defined. Independent and dependent variables. The independent variables are the impact of the risky events on the project time and cost which represented as the risks qualitative assessment using fuzzy multi-criteria risk analysis mode. While the dependent variables are the percentage of risk impact on the total project cost and duration. The system architecture consists of three layers. Input, middle and output layers. The number of the nodes located in the inputs and outputs layer is linked to the known and unknown facts of the problem. While the middle layer is determined based in the previous experience (historical data) (Haykin 2018). The ANN prediction model consists of two sets of groups. The first set is the training data set which used to the development of the model based on the historical data of achieved construction projects. While the second group is the testing data set which will be used to evaluate the performance of the ANN prediction mode. The Back-propagation (BP) method is chosen to train the applied artificial neutral network due to number of advantages such as high efficiency and simplicity. The outcomes after training the ANN is the prediction of each risk impact on the project cost and time.

5.5 Phase Five: Optimization Model for Selection the Optimal Risk Response Strategy

The most significant stage within the risk management process is risk response stage, where decision about the risky event is taken (Motaleb and Kishk 2012). This step designates strategy to manage a certain risk in the most efficient, appropriate and safe manner (Mubin and Mannan 2013). Risk response actions depend upon the type of risks if it a downs side risk (negative risk) or upside risk (opportunity). However, risk response planning stage is almost ignored part of the project risk management (Syedhosini et al. 2009). Literature investigation shows a limited research is carried out on the relation between risks and their response actions (Smith et al. 2014; Zhan 2016). According to most of the previous work in what concerning risk response actions have focused on human resources, experience and attitude factors and their relations to the binary variables (Cheraghi et al. 2017). Therefore, a mathematical model for selecting the optimal risk response strategies using Gravitational Search Algorithm (GSA) is proposed. The optimization model is to construct a mathematical model that considers the interaction between risks and relation between response actions in order to select an effective and efficient response action. The objective function in the model is to minimize the implementation costs of the response actions by taking into consideration time and cost constrains.

5.6 Phase Six: Risk Management Decision Support System

This phase will involve in the development of the proposed DSS computerized parts using (visual basic application software). This will include testing and validating the model to ensure its efficacy to the end users. To facilitate the process an interactive web-based decision support platform will be developed using the outcomes of the processed data. The online tool will be achieved using the following package:

(a) MySQL which is an open-source relational database management system. MySQL will be used for storing the primary data after being analysed. The stored data considered to be the core of the decision support system.
(b) The Hypertext Pre-processor (PhP) is a programming language together with Hypertext Mark-up Language 5 (HTML5) and Cascading Style Sheet (CSS) will be used for designing the user interface and functions of the decision support system.

According to Lam et al. (2007); the use of MySQL, PhP, HTML5 and CSS can provide the following benefits: the simplicity of using, data security, and the developer community is large in which it can extend their cooperation easily. They are open source content management programs. Therefore, it is not needed to pay for licenses as well as issues concern about the intellectual property when the decision support system comes into implementation. Therefore, using this package can provides a powerful decision support system development environment (Adam and Humphreys 2008). The designed DSS will enable decision makers to assess the oil and gas projects risky events; support their decision during planning and work implementation stages; gain experience in RM through exercising and implementing RM on scientific and documented bases; organize and document the knowledge-based information for subsequent generation of the oil and gas construction projects decision makers.

6 Conclusions

Oil and gas industry are an essential engine for economic in the developing countries. The construction projects in this sector are characterised as complex and risky due to their unique nature, dynamic environment, complex technology and numerous stakeholders. Oil and gas construction projects play a significant role in the sustainable supply chain and processing. Because of this importance, governments of developing countries are under pressure to ensure that construction projects will be completed on time within the estimated budget and the least amount of uncertainties and risks. However, the practice of construction risk management has not got the required attention in developing countries such as Iraq. The purpose of this research is to present an integrated methodology for managing construction risks in oil and gas fields in Iraq by focusing on analytical models to formalize and deal with human knowledge, risk impression and characteristics that are inherent. Also, to provide a clear differentiation between different risks and methods of understanding the interdependence

between risks. This paper concludes that establishing a structured methodology for managing construction projects risks for oil and gas projects will increase the effectiveness of projects risk management and improve project performance within the Iraqi oil and gas sector. In order to verify the proposed methodology, the next stage involves the use of documentary reviews, expert interviews and questionnaire survey for primary data collection (system requirement) and risk rating. Also, the feedback of the construction experts working in Iraqi oil and gas constriction projects will be used to validate the efficacy of the proposed research methodology. The design process of the risk analysis model, risk effect prediction and response action optimization model will be developed and tested based on real projects case studies in the mentioned area.

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